ClassAudit AI

Full Project Analysis Report

Face Recognition Attendance & Class Monitoring System

Report Date: 02 August 2026 • Version 1.0 • Confidential

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1. Executive Summary

ClassAudit AI is a production-grade, full-stack Django web application designed for educational institutions to automate teacher attendance tracking and live class monitoring using real-time facial recognition. The system delivers a **dual-portal role-based architecture** — one portal for **Principals** (administrators) and another for **Teachers** — each with dedicated dashboards, analytics, and management tooling.

The application integrates **computer vision** (OpenCV + dlib), **deep learning face embeddings** (128-dimensional vectors via the *face_recognition* library), and a **Retrieval-Augmented Generation (RAG) chatbot** backed by Groq's LLaMA 3.3 70B model for institutional document Q&A.

Django Apps	DB Models	API Routes	Templates	CSS Files	Source Lines
2	7	27	19	18	~3,500+

Key Highlights:

- Face recognition login with configurable Euclidean distance threshold (0.45)
- Multi-image enrollment (5 captures) with averaged 128-dim embeddings for high accuracy
- Live class monitoring with periodic face verification every 5 seconds
- RAG chatbot with per-principal ChromaDB vector store (PDF / MD / TXT ingestion)
- Defaulter CSV export filterable by department, day, month, and year
- Production-ready: WhiteNoise static serving, Gunicorn, HTTPS, HSTS, CSP headers

2. Project Overview

Attribute	Value
Project Name	ClassAudit AI — Smart Face Recognition Attendance System
Purpose	Automate teacher attendance and class monitoring for schools
Architecture	Monolithic Django MVC with dual-portal role-based access
Backend	Python 3.10+, Django 4.2.11 (LTS)
Database	SQLite3 (dev) / PostgreSQL (prod via psycopg2-binary)
Timezone	Asia/Kolkata (IST)
Authentication	Django session-based + face recognition webcam login
Deployment	Gunicorn + WhiteNoise; Procfile for PaaS (Heroku/Render)
Env Management	python-decouple (.env file — SECRET_KEY, GROQ_API_KEY, etc.)
License	Educational / Internal Use

Project Directory Structure

Path	Description
facere/	Django project config: settings, URLs, WSGI/ASGI
accounts/	Main app — all 27 views (1,661 lines), models, 19 templates
accounts/utils/	face_detector.py, face_embedding.py, face_matcher.py
chatbot/	RAG chatbot engine: rag_engine.py, views, URLs
static/	CSS (18 files), images
media/	Uploaded principal documents and face images
user_faces/	Teacher face photo uploads
data/users//	Per-teacher averaged face embedding (.npy files)
chroma_db/	ChromaDB vector store (per-principal collections)
docs/institutional/	Policy docs: 7 Markdown files (7 k–9 k chars each)
scripts/	Utility scripts (PDF generators, defaulter exporter)
requirements.txt	13 Python package dependencies
Procfile	Gunicorn deployment config
manage.py	Django management CLI entry point

3. Technology Stack

Layer	Technology / Version	Notes
Backend Framework	Django 4.2.11	LTS — stable, well-supported
Language	Python 3.10+	Required for dlib & face_recognition
Face Recognition	face_recognition 1.3.0	dlib ResNet-backed 128-dim embeddings
Computer Vision	OpenCV 4.8.1	Webcam capture, BGR-to-RGB conversion
Deep Learning	dlib 19.24.2	Facial landmark & encoding model
Numerics	NumPy 1.26.4	Embedding averages, distance math
Image Processing	Pillow 10.3.0	Image I/O and format handling
RAG Embeddings	sentence-transformers	all-MiniLM-L6-v2 (384-dim)
Vector Database	ChromaDB (persistent)	Per-principal cosine-similarity store
LLM (RAG)	Groq — LLaMA 3.3 70B	Fast inference via Groq Cloud API
Frontend	HTML5, CSS3, Vanilla JS	Modular per-page CSS, no JS framework
Charts	Chart.js	Department doughnut, teacher bar charts
Icons / Fonts	Font Awesome 6.4, Google Fonts	Plus Jakarta Sans, Bungee, Nunito
Static Serving	WhiteNoise 6.6.0	Compressed manifest static files
Production Server	Gunicorn 21.2.0	WSGI HTTP server
Database (prod)	PostgreSQL / psycopg2-binary	Switchable via DATABASE_URL
Env Config	python-decouple 3.8	.env-based secret management
PDF Generation	ReportLab	Institutional audit report PDFs

4. Architecture & Module Breakdown

ClassAudit AI uses a **monolithic Django MVC architecture**. All business logic resides in *accounts/views.py*, which acts as both controller and service layer. The RAG chatbot is isolated in a separate *chatbot* Django app with its own URL namespace. The face recognition pipeline is encapsulated in *accounts/utils/* as three single-responsibility modules.

4.1 Django Apps

App	Responsibility
accounts	Main application. Hosts all models, all 27 views, face recognition utilities, and 19 HTML templates. views.py is 1,661 lines covering both Principal and Teacher portals.
chatbot	RAG chatbot engine. Provides per-principal ChromaDB collections, document ingestion (PDF/MD/TXT), semantic retrieval, and Groq LLaMA answer generation. Exposed via /chatbot/ URL namespace.

4.2 Core Utility Modules (accounts/utils/)

Module	Responsibility
face_detector.py	Locates face bounding boxes in a webcam frame using face_recognition.face_locations().
face_embedding.py	Generates 128-dimensional face encodings. Accepts optional pre-detected bounding box. Returns None if zero or multiple faces found.
face_matcher.py	Compares a live embedding against a stored reference using Euclidean distance. Default threshold: 0.45.

4.3 Middleware Stack

Middleware	Purpose
SecurityMiddleware	HTTPS redirect, HSTS headers in production
WhiteNoiseMiddleware	Compressed static file serving without a CDN
SessionMiddleware	User session management
CommonMiddleware	URL normalisation
CsrfViewMiddleware	CSRF token enforcement on all POST requests
AuthenticationMiddleware	Attaches request.user on every request
MessageMiddleware	Django flash message framework
XFrameOptionsMiddleware	Clickjacking protection — X-Frame-Options: DENY (prod)

5. Database Models

The application defines **7 Django ORM models** stored in SQLite3 (dev) or PostgreSQL (prod). All primary keys use Django's default BigAutoField (64-bit integer).

Model	Purpose	Key Relation	Notable Fields
UserImages	Stores uploaded face photos per user	ForeignKey -> User	face_image (ImageField -> user_faces/)
Principal	School administrator account	OneToOne -> User	school_name
Teacher	Teacher profile linked to a school	ForeignKey -> Principal	name, department (13 choices)
Timetable	Weekly class schedule slot	ForeignKey -> Teacher	subject, day (MON-SAT), start_time, end_time
TeacherAttendance	Daily attendance record on check-in	ForeignKey -> Teacher	date (auto), time (auto), status
ClassSession	Tracks an active or completed class	FK -> Teacher + Timetable	total_active_duration, status, monitoring_resumption_count
PrincipalDocument	Institutional docs for RAG ingestion	ForeignKey -> Principal	file_type, uploaded_at, is_ingested

Signal Hooks: Two Django *post_delete* signals are registered. (1) On PrincipalDocument deletion — removes all corresponding ChromaDB vector chunks to keep the knowledge base clean. (2) On Teacher deletion — cascades to delete the associated Django User account and removes the face embedding directory from disk (*data/users/<username>/*).

6. API Route Inventory (27 Endpoints)

URL Pattern	Method	Access	Description
/	GET	Public	Landing page with live stats
/principal/register/	GET/POST	Public	Principal sign-up (AJAX JSON)
/principal/login/	GET/POST	Public	Principal password login
/principal/dashboard/	GET	Principal	Teacher overview + chatbot widget
/add-teacher/	GET/POST	Principal	Face-enrolled teacher registration
/principal/delete-teacher//	POST	Principal	Remove teacher + cascade face data
/principal/schedule//	GET/POST	Principal	Manage weekly timetable
/principal/delete-schedule//	POST	Principal	Delete single timetable slot
/principal/delete-all-schedule//	POST	Principal	Delete all timetable slots
/principal/reports//	GET	Principal	Attendance reports with filters
/principal/analysis/	GET	Principal	Dept charts, top-teacher analytics
/principal/teacher-analysis//	GET	Principal	Per-teacher deep analysis
/principal/export-defaulters/	GET	Principal	CSV download of absent teachers
/login/	GET/POST	Public	Face recognition webcam login
/teacher/login-password/	GET/POST	Public	Teacher username/password login
/teacher/dashboard/	GET	Teacher	Today's classes, calendar, history
/teacher/profile/	GET	Teacher	Monthly stats + class history
/teacher/mark-attendance/	GET/POST	Teacher	Face-based daily check-in
/teacher/start-class//	POST	Teacher	Initialises a ClassSession
/teacher/live-monitoring/	GET	Teacher	Real-time face tracking page
/teacher/update-live-attendance/	POST	Teacher	AJAX: periodic face check + duration
/teacher/end-class/	POST	Teacher	Finalise ClassSession
/teacher/records/	GET	Teacher	Filterable past session history
/teacher/help/	GET	Teacher	FAQ and support page
/logout/	GET	Auth	Session logout + redirect
/admin/	GET	Admin	Django admin interface
/chatbot/	POST	Auth	RAG chatbot API namespace

7. Feature Analysis

7.1 Principal Portal

Feature	Priority	Details
Dashboard	HIGH	Present/absent counts, teachers by dept, RAG chatbot, doc status
Teacher Management	HIGH	Add (5-image face capture), view, delete. Cascade removes user + face data
Timetable Scheduling	MEDIUM	CRUD for weekly slots per teacher (MON-SAT, subject, start/end time)
Attendance Reports	HIGH	Per-teacher history with date/month/year filter + Present/Late status
Performance Analysis	MEDIUM	Chart.js: dept distribution doughnut, top-teacher bars, active hours
Teacher Deep Analysis	MEDIUM	Per-teacher: attendance rate, avg active duration, session history
Defaulter CSV Export	MEDIUM	Filter by dept/day/month/year, download CSV of absent teachers
Document Management	HIGH	Upload PDF/MD/TXT -> per-principal ChromaDB for RAG Q&A;
RAG Chatbot	HIGH	Groq LLaMA 3.3 70B answers questions from institutional documents

7.2 Teacher Portal

Feature	Priority	Details
Dashboard	HIGH	Today's classes from timetable, class history, calendar widget
Profile	MEDIUM	Monthly attendance stats (Present/Late/Absent), class report history
Mark Attendance	HIGH	Face recognition webcam check-in; status: Present or Late
Start Class	HIGH	Launches a ClassSession linked to a timetable entry
Live Monitoring	HIGH	Face verified every 5 s; accumulates total_active_duration
End Class	HIGH	Finalises session, records duration and resumption count
Previous Records	MEDIUM	Filterable history of all past ClassSessions
Help Center	LOW	Static FAQ accordion page

8. Face Recognition Engine

The pipeline is built on the **face_recognition** library which wraps dlib's deep learning ResNet-based model. It operates in three distinct phases:

8.1 Enrollment (Teacher Registration)

- Principal captures **5 images** of the teacher via webcam during Add Teacher
- Each frame is converted from BGR to RGB (OpenCV default is BGR)
- 128-dim face embedding extracted per frame via face_recognition.face_encodings()
- All 5 embeddings averaged using np.mean(axis=0) to produce a single robust reference
- Averaged embedding saved as *data/users/<username>/embedding.npy*
- 5 raw images also stored in DB via the UserImages model

8.2 Authentication (Face Login)

- Teacher captures a webcam frame on the /login/ page via browser MediaDevices API
- Frame submitted as base64-encoded JPEG to the server
- Server decodes image, extracts 128-dim embedding, compares against all stored references
- Match confirmed if **Euclidean distance < 0.45**
- On match: corresponding Django User is retrieved and logged in via login(request, user)
- Exactly one face must be detected per frame — multi-face frames are rejected

8.3 Live Class Monitoring

- JavaScript polls /teacher/update-live-attendance/ every **5 seconds** with a frame
- Server performs face recognition on each received frame
- On successful match: elapsed interval is added to ClassSession.total_active_duration
- On failure (no face / mismatch): duration not incremented — teacher marked inactive
- Session resumes seamlessly if teacher returns; monitoring_resumption_count tracks re-entries
- On /teacher/end-class/: session finalized with status='Completed' and total duration saved

Parameter	Value
Embedding Model	dlib ResNet via face_recognition 1.3.0
Embedding Dimensions	128-dimensional Euclidean vector
Enrollment Images	5 per teacher, averaged into one reference
Match Threshold	Euclidean distance < 0.45
Live Poll Interval	5 seconds
Face Requirement	Exactly 1 face per frame for authentication/attendance

9. RAG Chatbot Engine

The chatbot is a full **Retrieval-Augmented Generation (RAG)** system for answering questions about institutional documents uploaded by each Principal. Every school has a completely isolated ChromaDB collection — a query from School A never touches School B's data.

Step	Description
1. Document Ingestion	Principal uploads PDF, MD, or TXT. Text is extracted (pypdf for PDF, direct read for MD/TXT), stripped of markdown formatting, and split into 600-character overlapping chunks (100-char overlap).
2. Embedding	Each chunk is embedded with all-MiniLM-L6-v2 (sentence-transformers, 384-dim). Stored in ChromaDB with metadata: title, doc_id, source_file, chunk_index.
3. Retrieval	On query, the question is embedded and top-5 closest chunks are retrieved via cosine similarity from the principal's ChromaDB collection.
4. Generation	Chunks are passed as context to Groq LLaMA 3.3 70B (temp=0.3, max_tokens=1024). A strict system prompt prevents hallucination — answers must originate from context.
5. Source Citation	Deduplicated document titles and relevance scores are returned alongside the answer.
6. Cleanup	Django post_delete signal on PrincipalDocument removes all associated ChromaDB chunks filtered by source_file.

Parameter	Value
Embedding Model	all-MiniLM-L6-v2 (sentence-transformers, 384-dim)
Vector Store	ChromaDB PersistentClient — cosine similarity space
LLM	Groq — llama-3.3-70b-versatile
Temperature	0.3 (factual, low creativity)
Max Tokens	1,024 per response
Chunk Size	600 characters with 100-character overlap
Retrieval Top-N	5 chunks per query
Supported Formats	PDF (pypdf), Markdown (.md), Plain Text (.txt)
Isolation Model	Per-principal ChromaDB collection: principal_

10. Code Metrics

10.1 Source Code Overview

File Type	File Count	Lines of Code
Python (.py)	35	4,873
HTML Templates (.html)	23	6,495
CSS Stylesheets (.css)	19	5,915
Total	77	17,283

10.2 Key File Analysis

File	Description	LOC	Functions
accounts/views.py	Main view layer — all business logic	1,660	27
chatbot/rag_engine.py	RAG pipeline: extract, chunk, embed, retrieve, generate	303	13
accounts/models.py	Database model definitions	139	9
facere/urls.py	URL routing — all 27 endpoints	51	0
facere/settings.py	Project settings (security, middleware, logging)	194	0
generate_pptx.py	Root-level PowerPoint presentation generator	433	3

11. Security Analysis

11.1 Security Strengths

- **Environment-based secrets:** SECRET_KEY and GROQ_API_KEY loaded via python-decouple — never hard-coded
- **CSRF protection:** CsrfViewMiddleware enabled on all POST endpoints
- **Role-based access control:** Custom @principal_required and @login_required decorators on all protected views
- **HTTPS enforcement:** SECURE_SSL_REDIRECT, SECURE_PROXY_SSL_HEADER, SESSION_COOKIE_SECURE, CSRF_COOKIE_SECURE (prod)
- **HSTS:** 1-year Strict-Transport-Security with includeSubDomains and preload in production
- **XSS protection:** SECURE_BROWSER_XSS_FILTER, SECURE_CONTENT_TYPE_NOSNIFF
- **Clickjacking protection:** X_FRAME_OPTIONS = 'DENY' in production
- **Password validation:** 4 Django validators (similarity, min-length, common passwords, numeric)
- **Input validation:** Alphanumeric username check, min-length enforcement, password complexity
- **Cascade deletion signals:** Face embeddings and user accounts removed automatically on teacher deletion
- **Logging:** Structured logging to console + app.log; DEBUG level in dev, INFO in prod

11.2 Security Concerns & Gaps

Severity	Issue	Detail
MEDIUM	2D Face Spoofing	No liveness detection — a printed photo could bypass webcam recognition
MEDIUM	No Rate Limiting	/login/, /mark-attendance/, /update-live-attendance/ have no brute-force protection
MEDIUM	SQLite3 in Production	settings.py defaults to SQLite; concurrent writes will cause DB lock errors
MEDIUM	Face Embeddings Unencrypted	data/users//embedding.npy stored as plaintext NumPy on disk
LOW	No Automated Tests	tests.py is empty; zero test coverage increases regression risk
LOW	ALLOWED_HOSTS Default	Defaults to localhost; production .env must override explicitly
LOW	ChromaDB No Auth	Local ChromaDB has no authentication — any local process can read vector data
INFO	views.py Monolith	1,661-line file increases maintenance burden and logic-bug surface area

12. Risks & Recommendations

ID	Severity	Risk	Recommendation
R1	HIGH	2D Face Spoofing — printed photo or video replay could bypass recognition	Implement liveness detection (blink/nod challenge, depth map, or anti-spoofing model)
R2	HIGH	Monolithic views.py — 1,661 lines of mixed logic makes testing and maintenance hard	Refactor into service classes or separate view modules per portal. Add unit tests.
R3	HIGH	No Rate Limiting — authentication endpoints vulnerable to brute-force attacks	Add django-axes or custom rate-limit middleware. Implement CAPTCHA after 5 failures.
R4	MEDIUM	SQLite3 in Production — not suited for concurrent writes or multi-worker deployments	Use dj-database-url and DATABASE_URL env var to switch to PostgreSQL in production.
R5	MEDIUM	GROQ_API_KEY Exposure Risk — key in process memory, risk if logs expose env vars	Rotate the API key regularly. Mask keys in logging. Ensure DEBUG=False in prod.
R6	MEDIUM	Unencrypted Face Embeddings on disk — data/users/ not encrypted at rest	Encrypt .npy files using Fernet (cryptography lib). Apply filesystem chmod 700.
R7	MEDIUM	Zero Test Coverage — tests.py is empty; regressions will go undetected	Write pytest/Django TestCase tests for all views, face matching, and RAG pipeline.
R8	LOW	ChromaDB has no authentication — any local process can read all vector data	For production: migrate to ChromaDB Cloud or Qdrant with API key authentication.
R9	LOW	AJAX endpoints rely on CSRF cookie — incorrect JS may silently fail	Confirm all AJAX POSTs include X-CSRFToken header. Add explicit @csrf_protect.
R10	LOW	ALLOWED_HOSTS not set for production — HTTP Host header attack risk	Explicitly set ALLOWED_HOSTS in .env for all environments.

13. Conclusion

ClassAudit AI is a **well-structured, feature-rich Django application** that successfully integrates cutting-edge AI capabilities — real-time face recognition and RAG-powered document Q&A — into a practical institutional management tool. The dual-portal architecture, modular CSS design system, and production-ready middleware stack demonstrate strong engineering fundamentals.

The system's most notable strengths are its **clean model-level separation of concerns**, robust production security configuration, and thoughtful UX across 19 dedicated templates and 18 modular stylesheets.

Priority improvements: (1) **liveness detection** to harden biometric security, (2) **decomposing views.py** into service modules, (3) **automated test coverage**, and (4) **rate limiting** on auth endpoints. Addressing these will elevate ClassAudit AI from a strong prototype to a fully production-audited institutional system.

Dimension	Assessment
Overall Quality	Production-ready prototype with a clear upgrade roadmap
Code Structure	Good model & utility separation; views.py needs decomposition
Security Posture	Strong prod config; face spoofing is the primary remaining gap
AI Integration	Excellent — both face recognition and RAG are well-implemented
Test Coverage	Critical gap — no automated tests exist
Deployment Readiness	Ready for PaaS (Heroku/Render) with minor .env configuration

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Report generated on 02 August 2026, 18:02 IST by Antigravity AI Code Analysis Engine.